Multi-armed bandits with arriving arms: sequential screening and reliable set maintenance

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Abstract

We study a stochastic multi-armed bandit problem in which new arms arrive exogenously and remain available. Performance is measured against the best currently available arm. The resulting arrival information discrepancy and drifting benchmark motivate UCB for Arriving Arms (UCB-AA), which screens new arms before they enter full competition with incumbents. We analyze this two-stage design and give a finite-budget regret analysis based on the comparisons and sample prefixes actually used. The analysis separates learning among admitted arms from delayed admission and loss of an earlier benchmark. We then ask what statistical benefit can accompany screening beyond a smaller internal active set. For simultaneous, reliable reporting of the relative near-optimal set, uncertainty about one arrival can obstruct many earlier certifications. We characterize a worst-schedule maintenance cost and identify the polynomial arrival boundary for sublinear unresolved time. A repeated-update family shows that prompt resolution of temporary membership questions can require inverse-square regret, whereas allowing more waiting admits a logarithmic upper bound. The necessary cost also holds with fixed positive probability. A separate checkpoint construction supplies attainable reporting bounds. Numerical evidence distinguishes the archived UCB-AA implementation from this construction and exposes the finite-budget cost of conservative certification.

Keywords: sequential experimentation; adaptive screening; arriving arms; dynamic regret; arrival information discrepancy; drifting benchmark.

1 Introduction

The classical multi-armed bandit framework assumes that the set of available actions is fixed in advance. In many modern applications, however, the action set expands over time. Precision oncology provides a representative example: as genomic profiling has advanced, the set of approved

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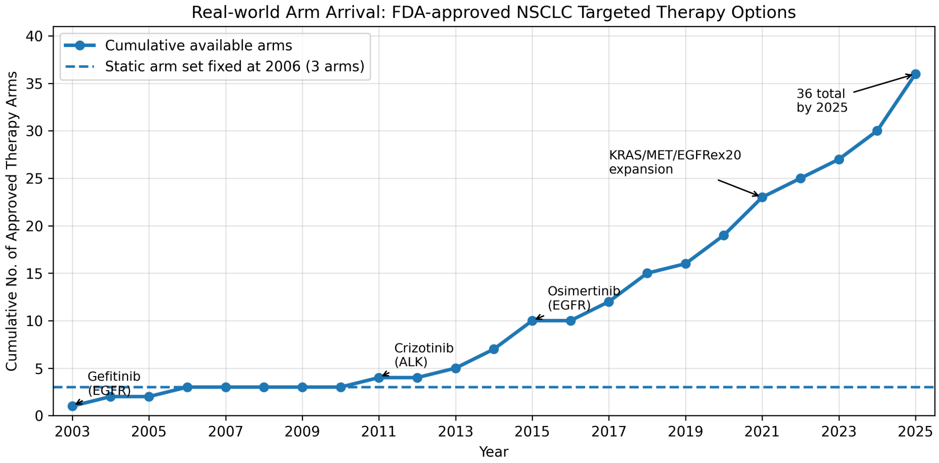

Figure 1: The historical motivating illustration from the original submission: FDA approval records for targeted therapies and immunotherapies for non-small cell lung cancer over 2003–2025 (U.S. Food and Drug Administration, 2025). The expanding option set motivates arriving-arm experiments. No treatment outcomes from this illustration enter the bandit simulations.

targeted therapies for non-small cell lung cancer has grown substantially, so a trial designed around a fixed collection of treatments may become outdated before it is completed. Similar issues arise in online advertising and other adaptive decision problems in which new options appear sequentially.

Motivated by such problems, we study the *multi-armed bandit problem with arriving arms* (MAB-AA). Let $\mathcal{A}_t$ denote the set of arms available at time t, with $\mathcal{A}_t \subseteq \mathcal{A}_{t+1}$. New arms arrive over time and remain available thereafter. Performance is evaluated against the best arm currently available, since a terminal best arm need not have been available earlier.

Relative to the classical stochastic bandit setting, MAB-AA introduces two main difficulties. The first is *arrival information discrepancy (AID)*: newly arrived arms must be explored, which repeatedly diverts samples away from well-understood incumbent arms. The second is a *drifting benchmark (DB)*: as better arms enter the system, both the identity of the optimal arm and the regret benchmark may change over time.

The setting retains stationary, unstructured reward laws and an external arrival process. Arms are not drawn on demand from a reservoir (Berry et al., 1997), and do not disappear as in mortal bandits (Chakrabarti et al., 2008). Permanent arrivals are a monotone special case of sleeping-bandit availability (Kleinberg et al., 2010). Benchmark changes are caused by new arms, rather than changes in the reward laws of existing arms as in nonstationary bandits (Besbes et al., 2014; Auer et al., 2019). Ballooning bandits already use permanent arrivals and regret against the best currently available arm (Ghalme et al., 2021); we do not claim either as a new model or comparator.

From a statistical perspective, multi-armed bandit problems have long been studied as models for sequential allocation and adaptive experimentation, dating back to Robbins (1952) and the asymptotically efficient allocation theory of Lai and Robbins (1985); see also Lai (1987) for connections to adaptive treatment allocation. This viewpoint has motivated a broad statistical literature, including Bayesian and index-based approaches to sequential allocation (Gittins, 1979; Berry and

Fristedt, 1985), response-adaptive randomization in clinical trials (Zelen, 1969; Wei and Durham, 1978; Villar et al., 2015; Williamson and Villar, 2020; Aziz et al., 2021), covariate-dependent and nonparametric bandit methods (Woodroofe, 1979; Yang and Zhu, 2002; Rigollet and Zeevi, 2010; Perchet and Rigollet, 2013), and recent high-dimensional, semiparametric, and transfer-learning formulations (Qian and Yang, 2016; Qian et al., 2024; Cai et al., 2024). These works share with ours the broad goal of learning from accumulated evidence to reduce allocation to inferior alternatives, but do not directly study the permanent exogenous arrival setting considered here.

Methodologically, the two challenges call for different but complementary design principles. To address AID, our approach combines elimination steps (Even-Dar et al., 2006; Auer and Ortner, 2010; Perchet and Rigollet, 2013; Qian and Yang, 2016) with a pre-elimination mechanism tailored to the expanding action set. The key idea is to prevent the main competition stage from being repeatedly disrupted by every newly arriving arm. Instead, late arrivals are first screened against incumbent candidates, so that clearly inferior arms can be removed before receiving the full exploration budget assigned to serious contenders. To handle DB, the algorithm repeatedly updates the relevant comparison set as new arms arrive, and the analysis evaluates regret against the best arm currently available rather than against a fixed hindsight oracle. Based on these ideas, we propose **UCB for Arriving Arms (UCB-AA)**. For UCB-AA, we derive regret upper bounds that depend explicitly on the arrival process, establish sublinear dynamic regret when arm-count growth and admission delay are controlled, study representative lower-bound regimes, and develop an online extension for unknown horizons.

UCB for Arriving Arms (UCB-AA) implements this screening design. Its pre-elimination stage samples an arrival at coarse resolutions before spending the finer budget used for incumbent arms. Formal elimination then compares the merged active set. A finite-budget analysis must account for unfinished phases, arrivals waiting for admission, and any loss of a previously admitted best arm. We give that accounting explicitly and establish a sample-count bound without compressing historical gaps into a terminal gap.

The active set alone does not describe what has been learned about all available arms. An excluded arm may be statistically inferior, or it may never have been admitted. This distinction leads to a further question within the same MAB-AA setting: what does it cost to maintain reliable knowledge of the complete relative near-optimal set while collecting reward? The fixed-arm version is the ALL-ε problem (Mason et al., 2020). Joint reward and identification objectives also have established precedents (Degenne et al., 2019; Yang et al., 2026; Cho et al., 2025). Our additional requirement concerns continuing reports under arrivals, with one simultaneous correctness event and a cumulative cost for unresolved arm time.

Two effects of the drifting benchmark determine that cost. An uncertain new arm can block positive certification of many well-studied incumbents. A later improvement can also make a difficult earlier membership question irrelevant. We prove a common-environment broadcast lower bound and a worst-schedule maintenance bound, obtaining the polynomial arrival exponent below which sublinear regret and sublinear unresolved time are jointly possible. In a repeated-update family, prompt maintenance has matching inverse-square regret bounds at specified horizons, whereas a larger constant-factor area budget admits a logarithmic regret upper bound. A fixed-probability lower bound shows that the necessary cost is not confined to rare paths. These are substantive extensions of the original regret question, rather than guarantees inferred from the size of UCB-AA's active set.

For the reporting upper bounds we use a separate checkpoint construction, CP-UCB-AA. It preserves accumulated observations but does not remove negatively reported arms from reward selection. We distinguish its guarantees from those of the two-stage UCB-AA procedure. The numerical study likewise distinguishes the original archived implementation from the reporting comparisons. It retains favorable and unfavorable original observations, while showing that conservative fixed checkpoints can be impractical at moderate budgets.

2 Multi-armed bandit problem with arriving arms

2.1 Problem formulation

We consider a sequential decision process over time steps $t = 1, 2, \ldots$. Let $\mathcal{A}$ be the possibly infinite set of all arms. At time t only $\mathcal{A}_t \subseteq \mathcal{A}$ is available, where $\mathcal{A}_t \subseteq \mathcal{A}_{t+1}$, $\mathcal{A}_1 \neq \varnothing$, and $K_t = |\mathcal{A}_t| < \infty$. For arm i, its arrival time is

$$\tau_i = \inf\{t \geq 1 : i \in \mathcal{A}_t\}.$$

Each arm has an unknown stationary reward law on $[0, 1]$ with mean r_i. Reward streams are independent across arms, and successive observations of each arm are independent and identically distributed. The arrival process is exogenous to the learner and independent of these reward streams. We state results conditional on a fixed arrival schedule unless otherwise specified.

In physical-time notation, $X_{t,i}$ denotes the reward observed if arm i is selected at time t. A policy is a sequence $\pi = (\pi_t)_{t\geq 1}$ of possibly randomized rules. After seeing $\mathcal{A}_t$, it selects $I_t \in \mathcal{A}_t$ using the observed history

$$H_{t-1} = (\mathcal{A}_s, I_s, X_{s,I_s})_{s=1}^{t-1}$$

and independent randomization, and then observes X_{t,I_t}. It does not know the arm means or future arrivals.

Write $N_i(t) = \sum_{s\leq t} \mathbf{1}\{I_s = i\}$ for the cumulative number of pulls of arm i. To distinguish pull number from physical time, represent its successive potential observations by $(Y_{i,n})_{n\geq 1}$, so that $X_{t,I_t} = Y_{I_t, N_{I_t}(t-1)+1}$. Define

$$\hat{r}_i(n) = \frac{1}{n}\sum_{s=1}^{n} Y_{i,s}, \qquad r^*(t) = \max_{i\in\mathcal{A}_t} r_i, \qquad \Delta_i(t) = r^*(t) - r_i \quad (i \in \mathcal{A}_t).$$

For an almost surely finite stopping time τ, the cumulative dynamic regret is

$$R_\tau(\pi) = \sum_{t=1}^{\tau} \bigl(r^*(t) - r_{I_t}\bigr).$$

At a deterministic horizon T we write $R_T(\pi)$, or R_T when the policy is understood. It is random through the arm selections; $\mathbb{E}R_T$ is its expectation. A definition at random τ does not assert that every deterministic-horizon bound also holds at arbitrary stopping times. Physical time t always counts actual pulls. Algorithmic rounds are indexed separately by m.

2.2 Main statistical challenges

Arrival information discrepancy (AID). A first difficulty is arrival information discrepancy. With a finite sampling budget up to time T, the learner must allocate samples both to incumbent arms and to newly arriving arms whose rewards are still highly uncertain. Under continual arrivals, this repeated re-exploration can prevent the learner from concentrating pulls on the best currently available arm. Consequently, without additional regularity conditions on the arrival process, sublinear regret may be unattainable.

Drifting benchmark (DB). In contrast to the classical setting where the best mean reward is fixed, the benchmark in MAB-AA is time-dependent: $r^*(t) = \max_{i \in \mathcal{A}_t} r_i$. Since $\mathcal{A}_t$ expands over time, both the identity of the best arm and the value $r^*(t)$ may change when new arms arrive. Consequently, for a fixed arm i, the suboptimal gap $r^*(t) - r_i$ is generally time-varying (and non-decreasing in t). This drifting benchmark complicates both algorithm design and analysis, because the learner must continuously compare well-sampled incumbent arms with newly arriving arms whose estimates are still highly uncertain.

2.3 Admission and retirement in the reward account

The distinction between statistical reports and reward eligibility is useful even before introducing a policy. Let $\mathcal{A}_t^{\mathrm{adm}} \subseteq \mathcal{A}_t$ be an admitted set and $B_t \subseteq \mathcal{A}_t^{\mathrm{adm}}$ its nonempty reward-active subset, with $I_t \in B_t$. At every deterministic endpoint, including one inside an algorithmic round,

$$\begin{aligned} R_T = {} & \sum_{t \le T} (\max_{i \in B_t} r_i - r_{I_t}) + \sum_{t \le T} (r^*(t) - \max_{i \in \mathcal{A}_t^{\mathrm{adm}}} r_i) \\ & + \sum_{t \le T} (\max_{i \in \mathcal{A}_t^{\mathrm{adm}}} r_i - \max_{i \in B_t} r_i). \end{aligned} \tag{1}$$

This follows by adding and subtracting the two intermediate maxima at each pull. The terms are within-active learning, admission delay, and loss of an admitted benchmark through retirement. The latter two cannot in general be identified with arrival increments alone. On a simultaneous confidence event, the revised UCB-AA comparisons preserve a best admitted arm, making the third term zero. The checkpoint construction introduced later uses $B_t = \mathcal{A}_t^{\mathrm{adm}} = \mathcal{A}_t$ for reward selection, so its last two terms vanish on every sample path.

3 UCB for Arriving Arms (UCB-AA)

UCB-AA is an elimination-based procedure organized by rounds. Its main idea is simple: newly arriving arms are not allowed to compete immediately with well-sampled incumbent arms. Instead, they are first buffered and screened at lower cost. Only those that survive this screening step enter full competition in the next stage. This design aims to reduce wasted exploration on clearly suboptimal late arrivals while preserving the ability to detect genuinely competitive new arms.

3.1 UCB-AA under a physical sampling budget

UCB for Arriving Arms (UCB-AA) uses two stages of screening. At the beginning of round m, admit the buffered batch A_m and form $B_m = C_{m-1} \cup A_m$, where C_{m-1} contains the surviving incumbents. For $m \geq 2$, screen the new arms in phases $p = 0, \ldots, m-2$ against the incumbents and the remaining new arms; only new arms can be removed. Then apply formal elimination to the merged set in phases $m-1, m$. Round zero uses phase zero, and round one uses formal phases zero and one. Empty preliminary batches are no-ops. This retains the distinction between pre-elimination, which addresses arrival information discrepancy, and formal elimination after new arms have entered the main competition.

Let s_m be the physical start of round m. Set $A_0 = \mathcal{A}_1$ and, for $m \geq 1$, admit the entire buffer, including arrivals at s_m: $A_m = \mathcal{A}_{s_m} \setminus A(m-1)$, where $A(m-1) = \bigcup_{j<m} A_j$. Arrivals during a call are added to the buffer before the current physical action and wait for the next round.

To give this procedure a fixed-budget guarantee, specify its confidence calibration, comparison data, and stopping rule. For a known physical budget $T \geq 1$ and $0 < \delta < 1$, let

$$M_T = \left\lceil \tfrac{1}{2} \log_2(T+1) \right\rceil, \qquad N = 8T4^{M_T}/\delta, \qquad \widetilde{\Delta}_p = 2^{-p}. \tag{2}$$

Use the target and radius formulas

$$n_p = \left\lceil \frac{2\log(N\widetilde{\Delta}_p^2)}{\widetilde{\Delta}_p^2} \right\rceil, \qquad b_p = \sqrt{\frac{\log(N\widetilde{\Delta}_p^2)}{2n_p}}. \tag{3}$$

In an elimination call, sample each candidate to n_p lifetime pulls, retaining all earlier observations. After the call's sampling is complete, compare the actual prefix intervals $\hat{r}_i(n_p) \pm b_p$. Remove a candidate only if its upper bound is strictly below the largest reference lower bound. Incumbents have already reached the necessary prefix before a preliminary comparison, so no unobserved incumbent estimate is used. Stop after physical pull T, including when it cuts through a target or a round. The calibrated $n_{M_T} > T$ ensures that the finite phase range covers this execution. These conventions specify the confidence comparisons and budget accounting for the two elimination stages. The guarantee below uses this calibration; a smaller empirical calibration need not have the same coverage probability.

The algorithm retains the original two stages. Pre-elimination applies only to newly admitted arms and uses the already observed incumbent prefixes. Formal elimination applies to the merged active set. The empty-batch branch, strict comparison, and physical-budget stopping rule make those operations unambiguous; they do not replace screening with the checkpoint construction.

Algorithm 1 UCB for Arriving Arms (UCB-AA), with explicit prefix comparisons

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Require: Physical budget $T$; $M_T, N, n_p, b_p$ from (2)–(3)
Initialize $C_{-1} = \varnothing$ and retain lifetime sample prefixes
Stop immediately after pull $T$; an unfinished call performs no elimination
procedure ELIM($S, R, p$)
    Before each physical pull, add newly arriving unadmitted arms to the buffer
    Pull each $j \in S$ to $n_p$ total observations
    For $j \in R$, set $\mathrm{LCB}_j = \hat{r}_j(n_p) - b_p$, $\mathrm{UCB}_j = \hat{r}_j(n_p) + b_p$
    Remove simultaneously from $S$ every $j$ with $\mathrm{UCB}_j < \max_{i \in R} \mathrm{LCB}_i$
    return $S$
end procedure
for $m = 0, \dots, M_T$ do
    Admit buffered batch $A_m$; set $B_m^0 = C_{m-1} \cup A_m$
    if $m \geq 2$ then
        Set $A_m^0 = A_m$
        for $p = 0, \dots, m-2$ do
            if $A_m^p \neq \varnothing$ then
                $A_m^{p+1} \leftarrow$ ELIM($A_m^p, C_{m-1} \cup A_m^p, p$)
            else
                $A_m^{p+1} \leftarrow \varnothing$
            end if
            $B_m^{p+1} \leftarrow C_{m-1} \cup A_m^{p+1}$
        end for
    end if
    if $m = 0$ then
        $B_m^1 \leftarrow$ ELIM($B_m^0, B_m^0, 0$)
    else
        for $p = m-1, m$ do
            $B_m^{p+1} \leftarrow$ ELIM($B_m^p, B_m^p, p$)
        end for
    end if
    $C_m \leftarrow B_m^{m+1}$
end for
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3.2 Roundwise regret, historical loss, and arrival delay

Let s_m be the physical start of round m, $A(m) = \bigcup_{j \leq m} A_j$, and $r_{*(m)} = \max_{i \in B_m} r_i$. Write $\Delta_i^{\mathrm{rd}}(m) = r_{*(m)} - r_i$ for its roundwise gap, reserving $\Delta_i(t) = r^*(t) - r_i$ for physical time. Write $r_{\mathrm{adm}}(m) = \max_{i \in A(m)} r_i$ for the best mean among all admitted arms, including those previously eliminated. At a physical time t, the current round is $m(t)$. The exact regret identity is

$$\begin{aligned} R_T(\pi) &= R_{\mathrm{round}}(T) + \mathcal{H}_T + D_T, \qquad (4)\\ R_{\mathrm{round}}(T) &= \sum_{t \leq T} \{ r_{*(m(t))} - r_{I_t} \}, \\ \mathcal{H}_T &= \sum_{t \leq T} \{ r_{\mathrm{adm}}(m(t)) - r_{*(m(t))} \}, \\ D_T &= \sum_{t \leq T} \{ r^*(t) - r_{\mathrm{adm}}(m(t)) \}. \end{aligned}$$

All three quantities are nonnegative. The historical term $\mathcal{H}_T$ records the consequence of eliminating an admitted benchmark arm; D_T records only the time before arrivals are admitted. Combining them into a residual does not make that residual arrival delay.

For a complete run of $M+1$ rounds, the original $T_0(\pi)$ denotes its actual random number of pulls, and $R_{\text{round}}(T_0(\pi)) = R_M(\pi)$. The pathwise identity also applies at this endpoint, but the fixed-budget bound does not replace T by $\mathbb{E}T_0(\pi)$ or assert a guarantee for an arbitrary stopping time. The original $K(m)$, if supplied, bounds $|A(m)|$; it differs from the actual count $K_t = |\mathcal{A}_t|$ and is not an additional input required by the present calibration.

The delay contribution is explicit. Let $\beta_u = r^*(u) - r^*(u-1)$ for $u \geq 2$ and let $\alpha_T(u)$ be the first round start at or after u, with value $T+1$ when no such start occurs by T. Because $A(m) = \mathcal{A}_{s_m}$,

$$D_T = \sum_{u=2}^{T} \beta_u \{\alpha_T(u) - u\}. \tag{5}$$

This formula retains benchmark improvements in the last unfinished round.

Theorem 3.1 (A repaired fixed-budget bound for UCB-AA)**.** *For independent arm streams, each consisting of iid* $[0,1]$*-valued rewards, deterministic permanent arrivals, a nonempty initial arm set, and* $K_T = |\mathcal{A}_T| < \infty$*, the UCB-AA procedure specified above satisfies, with probability at least* $1-\delta$*,*

$$\mathcal{H}_T = 0, \qquad R_T(\pi) \leq \min\{T,\ 8\sqrt{(2\log N + 1)K_T T} + D_T\}.$$

It follows that

$$\mathbb{E}R_T(\pi) \leq \min\{T,\ 8\sqrt{(2\log N + 1)K_T T} + \mathbb{E}D_T + \delta T\}. \tag{6}$$

The inequalities include incomplete targets and rounds and allow ties between arm means. No terminal-gap stability condition is needed.

The proof uses the actual lifetime observation count. On one simultaneous prefix-coverage event, a best admitted arm survives every comparison. A new arm that receives phase-p samples for $p \geq 1$ has survived the preceding comparison in the same round; its roundwise gap is at most $4b_{p-1}$. An incumbent receives no new samples in formal phase $m-1$, because it already has n_{m-1} observations. Its new phase-m samples follow the current round's phase-$m-1$ comparison and obey the same bound. Thus the nth observation of an arm has roundwise loss at most $4\sqrt{(2\log N + 1)/n}$. Summing these distinct prefix charges yields (6); the argument does not replace intermediate gaps by a terminal gap or condition on a selected elimination chain. The supplementary material gives the full proofs.

For $T \geq 2$, the choice $\delta = T^{-2}$ makes the statistical term $O(\sqrt{K_T T \log(T+1)})$. Sublinear regret therefore follows under the sufficient conditions $K_T \log(T+1) = o(T)$ and $\mathbb{E}D_T = o(T)$. The delay condition is essential to this statement; the result does not assert sublinear regret for arbitrary arrivals. For an unknown horizon, a restart at physical budgets $1, 2, 4, \ldots$ gives a valid extension with the same statistical order and its own explicit delay term. The supplementary material provides its construction. Each restart has a prescribed number of physical pulls, so an unfinished round is charged at the actual endpoint.

Corollary 3.2 (Static arms and fixed-gap distractors)**.** *Suppose an arm available at time one has mean $r^\star$ and every arm arriving by T has mean at most $r^\star$. Write $\Delta_i = r^\star - r_i$ and $A = 2\log N + 1$. Then $D_T = 0$ pathwise and, with probability at least $1-\delta$,*

$$R_T(\pi) \le \sum_{\substack{i\in\mathcal{A}_T\\ \Delta_i>0}} \min\left\{\Delta_i(T-\tau_i+1), \frac{16A}{\Delta_i}\right\}. \tag{7}$$

Adding δT gives an expected-regret bound, and both bounds may be capped at T.

Indeed the active best mean equals $r^\star$ on the common confidence event. Applying the actual-prefix charge to a suboptimal arm's last observation gives $N_i(T) \le 16A/\Delta_i^2$, while $N_i(T) \le T - \tau_i + 1$. For $T \ge 2$ and $\delta = T^{-2}$, the static case therefore has upper order $\sum_{i:\Delta_i>0} \log(T+1)/\Delta_i$; arrivals with a fixed positive gap have upper order $K_T \log(T+1)$. For example, consider an initial optimum $r^\star = 3/4$, another initial mean $1/4$, and arriving means $r_i = 1/2 - 2^{-i-2}$ at exogenous physical times τ_i, $i \ge 1$. All distractors have gap at least $1/4$, so the same bound gives $O(K_T \log(T+1))$ expected regret for this exogenous arrival example.

4 Regret lower bounds in representative arrival regimes

We consider two representative lower-bound regimes for regret. The first describes the evidence needed to reject inferior arrivals when the best arm is already available. Its static special case is a classical calibration (Lai and Robbins, 1985; Burnetas and Katehakis, 1996). The second describes successive local comparisons as the benchmark improves. Both concern regret at a deterministic physical horizon. They are separate from the reporting requirements introduced below, and are not claimed as matching converses to an upper bound evaluated at a random round-completion time.

4.1 Static arms and arriving distractors

For each horizon T, fix an exogenous deterministic arrival schedule and a baseline environment $\nu^{(0)}$. Arm 0 is available from time 1 and has mean $r^\star$ strictly larger than every other mean. Let $\mathcal{I}_T$ be a nonempty collection of inferior arrivals. For $i \in \mathcal{I}_T$, write

$$r_i = r^\star - \Delta_i, \qquad H_i(T) = T - \tau_i + 1, \qquad H_{\min}(T) = \min_{i\in\mathcal{I}_T} H_i(T).$$

The reward law of arm i is P_i. Suppose there is a law Q_i on $[0,1]$ with mean $r^\star + \eta_i$, where $\eta_i > 0$, and put $d_i = D(P_i \| Q_i)$. Let $\nu^{(i)}$ change only this arm's law to Q_i, retaining all other reward laws and the arrival schedule. Set $\mathcal{V}_T = \{\nu^{(0)}\} \cup \{\nu^{(i)} : i \in \mathcal{I}_T\}$.

Theorem 4.1 (Local-alternative regret lower bound)**.** *Assume $H_{\min}(T) \to \infty$ and that fixed positive constants $\underline{\Delta}, \underline{\eta}, \overline{d}$ satisfy, for all sufficiently large T and every $i \in \mathcal{I}_T$,*

$$\Delta_i \ge \underline{\Delta}, \qquad \eta_i \ge \underline{\eta}, \qquad 0 < d_i \le \overline{d}.$$

Suppose a sequence of nonanticipating policies $\{\pi^{(T)}\}$ *is uniformly good on this family at scale* $H_{\min}(T)$*, meaning*

$$\sup_{\nu \in \mathcal{V}_T} \mathbb{E}_\nu R_T(\pi^{(T)}; \nu) = o\big(H_{\min}(T)^a\big) \quad \textit{for every } a > 0. \tag{8}$$

Then, for all sufficiently large T*,*

$$\mathbb{E}_{\nu^{(0)}} R_T(\pi^{(T)}; \nu^{(0)}) \geq \frac{1}{12} \sum_{i \in \mathcal{I}_T} \frac{\Delta_i}{d_i} \log H_i(T) \geq \frac{\underline{\Delta}}{12\bar{d}} |\mathcal{I}_T| \log H_{\min}(T). \tag{9}$$

The conclusion is conditional on (8); that condition can be restrictive for growing families. Indeed, the theorem requires $|\mathcal{I}_T| \log H_{\min}(T) = o(H_{\min}(T)^a)$ for every $a > 0$ if such a policy sequence exists. It does not assert existence in arrival regimes that violate this necessary compatibility condition. Equally spaced arrivals with spacing $\lfloor T/|\mathcal{I}_T| \rfloor \to \infty$ give $H_{\min}(T) \geq \lfloor T/|\mathcal{I}_T| \rfloor$. If every arrival occurs before αT for a fixed $\alpha < 1$, then $H_{\min}(T) \geq (1-\alpha)T$. These substitutions describe the corresponding lower bounds without removing the uniform-goodness condition.

For a fixed collection of K inferior arms all available at time 1, take $H_i(T) = T$. If the local alternatives satisfy $d_i \leq C_i \Delta_i^2$ with fixed $0 < C_i < \infty$, then the same argument gives

$$\mathbb{E}_{\nu^{(0)}} R_T(\pi^{(T)}; \nu^{(0)}) \geq \frac{1}{12} \sum_{i=1}^{K} \frac{\log T}{C_i \Delta_i} \tag{10}$$

for all sufficiently large T. The laws, gaps, and K are fixed in this specialization. These are applications of the usual bandit change-of-measure argument (Kaufmann et al., 2016), not new claims about a regret comparator. Complete proofs appear in the supplementary material.

4.2 Successive near-tied benchmark improvements

Fix integers $J \geq 1$ and $H_1, \ldots, H_J \geq 1$ with $\sum_{m=1}^{J} H_m = T$. Write $s_0 = 0$, $s_m = \sum_{\ell=1}^{m} H_\ell$, and $\mathcal{B}_m = \{s_{m-1}+1, \ldots, s_m\}$. At the start of block m, two Bernoulli arms c_m, d_m arrive and remain available, with means

$$r_{c_m} = \theta_m, \qquad r_{d_m} = \theta_m + \sigma_m \varepsilon_m, \qquad \sigma_m \in \{-1, +1\}.$$

There are no other arms in this construction, so $K_T = 2J$. Fix $0 < \gamma < 1/2$ and require

$$\theta_m - \varepsilon_m,\ \theta_m + \varepsilon_m \in [\gamma, 1-\gamma], \qquad \varepsilon_m > 0, \qquad \theta_{m+1} \geq \theta_m + \varepsilon_m \quad (m < J). \tag{11}$$

Let $\mathcal{V}_T^{\text{prog}}$ contain the 2^J sign configurations. The local perturbations ε_m in this construction are not the near-optimality tolerance used later for reporting.

Theorem 4.2 (Deterministic-block regret lower bound)**.** *Every nonanticipating policy satisfies*

$$\sup_{\nu\in\mathcal{V}_T^{\mathrm{prog}}} \mathbb{E}_\nu R_T(\pi;\nu) \geq \frac{1}{2}\sum_{m=1}^{J} H_m\varepsilon_m \left(1-\sqrt{\frac{2H_m\varepsilon_m^2}{\gamma(1-\gamma)}}\right)_+ . \tag{12}$$

In particular, if $H_m\varepsilon_m^2 \leq \kappa\gamma(1-\gamma)$ *for all* m*, with* $0<\kappa<1/2$*, the right side is at least* $c_\kappa \sum_m H_m\varepsilon_m$*, where* $c_\kappa = (1-\sqrt{2\kappa})/2 > 0$*. The same bounds hold after taking the infimum over policies on the left.*

If $H_m \asymp T/J$, $\varepsilon_m \asymp \sqrt{J/T}$, and the constants satisfy the preceding small-information condition, this gives a lower bound of order $\sqrt{JT}$. Bounded rewards restrict this specialization. If $\varepsilon_m \geq c_\varepsilon\sqrt{J/T}$ for every m, then

$$Jc_\varepsilon\sqrt{J/T} \leq \sum_{m=1}^{J}\varepsilon_m \leq \theta_J + \varepsilon_J - (\theta_1 - \varepsilon_1) \leq 1-2\gamma.$$

Consequently $J = O(T^{1/3})$ is necessary for this family. This is a feasibility restriction of the bounded ladder construction, not an impossibility boundary for the unrestricted arriving-arm model. No uniform-goodness assumption is needed for (12). Complete proofs appear in the supplementary material.

5 Reliable maintenance of the near-optimal set

We now supplement the allocation objective with an explicit reporting objective. The reward model, available arms, and dynamic-regret comparator remain those of Section 2. Reporting a relative near-optimal set is an additional statistical task; it is not identified with UCB-AA's active or surviving set. We use separate report notation to keep that distinction visible.

5.1 Reporting reliability and unresolved time

For $0<\varepsilon<1$, the target is

$$\mathcal{G}_t^\varepsilon = \{i\in\mathcal{A}_t : \Delta_i(t)\leq\varepsilon\}.$$

A report partitions $\mathcal{A}_t$ into $\mathcal{P}_t, \mathcal{N}_t, \mathcal{V}_t$: certified good, certified bad, and unresolved. With $0<\delta<1/2$, let

$$E_{\mathrm{lab}} = \{\text{for all } t:\ \mathcal{P}_t \subseteq \mathcal{G}_t^\varepsilon \subseteq \mathcal{A}_t\setminus\mathcal{N}_t\}.$$

A policy is δ-reliable if $\mathbb{P}(E_{\mathrm{lab}}) \geq 1-\delta$ in every allowed reward environment. Positive reports may need revision after an arrival. A sound negative report remains valid because $r^*(t)$ is nondecreasing. This statement refers to the model's stationary means and permanent availability.

Reliability alone permits every arm to remain unresolved. To measure progress, fix a boundary width $0<\eta<1$ and define

$$d_i^\varepsilon(t) = |\Delta_i(t)-\varepsilon|, \quad U_t(\eta) = \sum_{i\in\mathcal{A}_t}\mathbf{1}\{i\in\mathcal{V}_t, d_i^\varepsilon(t)>\eta\}, \quad \mathcal{U}_T(\eta) = \sum_{t=1}^{T}U_t(\eta). \tag{13}$$

An arm within the boundary band still requires a correct label whenever a label is issued; its unresolved time is simply exempt from the service loss. No assumption removes such arms from the model. The parameters have different roles: ε defines membership, η defines the precision at which unresolved time is charged, and a policy's service precision will be denoted by ρ.

The main joint requirement at a fixed T is

$$\mathbb{E}_\theta R_T \le r, \qquad \mathbb{P}_\theta\{E_{\mathrm{lab}}, \mathcal{U}_T(\eta) \le a\} \ge 1 - \delta. \tag{14}$$

It uses an expected reward cost and a joint high-probability reporting cost. It is stronger than control of per-arm errors, and differs from false discovery rate control. In particular, $\mathcal{U}_T = o(T)$ requires the time-average *number* of unresolved separated arms to vanish. It does not merely require a small unresolved fraction among all arrived arms.

For a performance family Θ, write $\mathfrak{R}_\Theta(a)$ for the infimum of $\sup_{\theta\in\Theta} \mathbb{E}_\theta R_T$ over policies reliable on the full bounded model and satisfying the second condition of (14) for every $\theta \in \Theta$. The value is $+\infty$ if no policy is feasible. The lower bounds remain valid if reliability is required only on the lower-bound subfamily.

5.2 Lifetime checkpoints and reward selection

For this construction, index arms by arrival order $i = 1, 2, \ldots$, with a fixed order within each batch. This is an indexing convention, not an assumption about the reward laws. Choose $0 < \rho < 1$ before observing rewards. For arm i, set

$$q_i(\rho) = \left\lceil \frac{2}{\rho^2} \log \frac{4i(i+1)}{\delta} \right\rceil. \tag{15}$$

Before this count is reached, its reporting interval is $[0, 1]$. Afterwards, the interval is permanently

$$[\ell_i, u_i] = [\hat{r}_i(q_i) - \rho/2, \hat{r}_i(q_i) + \rho/2] \cap [0, 1]. \tag{16}$$

The interval uses a fixed potential prefix, even though its completion time is adaptive. Later observations continue to update the reward estimate. One can also prescribe different precisions ρ_i before observing the corresponding arms; selecting a favorable checkpoint from their realized data is not covered.

CP-UCB-AA uses the following physical-clock rule.

1. Admit current arrivals and append their checkpoints to a queue in arrival order. Initialize their counts to zero.
2. At an even time step, if a checkpoint is incomplete, sample the first incomplete arm in the queue. These are designated service pulls.

3. At every other time step maximize, over all $i \in \mathcal{A}_t$,

$$J_i(t) = \hat{r}_i(N_i(t-1)) + \sqrt{\frac{3\log(2t)}{N_i(t-1)}}. \tag{17}$$

   Use $+\infty$ for a zero count and break ties in arrival order.

4. Update the chosen arm's lifetime count and reward sum. If the count first equals q_i, construct (16). Every observation contributes to checkpoint completion, including observations from (17).

5. Set $\underline{r}^*(t) = \max_{i\in\mathcal{A}_t} \ell_i$ and $\overline{r}^*(t) = \max_{i\in\mathcal{A}_t} u_i$, and report

$$\mathcal{P}_t = \{i : \ell_i \geq \overline{r}^*(t) - \varepsilon\}, \qquad \mathcal{N}_t = \{i : u_i < \underline{r}^*(t) - \varepsilon\}, \qquad \mathcal{V}_t = \mathcal{A}_t \setminus (\mathcal{P}_t \cup \mathcal{N}_t). \tag{18}$$

The intervals are nonempty on every sample path, so $\mathcal{P}_t$ and $\mathcal{N}_t$ cannot overlap. All arrived arms remain eligible for (17), including those in $\mathcal{N}_t$. The method stores a count, sum, and reporting interval for every arm; it does not claim to reduce total arm storage. A direct index or report update costs $O(K_t)$ operations. The executable fixed-precision reference is supplied with the numerical code.

Proposition 5.1 (Simultaneous reporting and unconditional reward cost)**.** *On the full bounded model, CP-UCB-AA satisfies $\mathbb{P}(E_{\text{lab}}) \geq 1 - \delta/2$. If $\mathcal{I}_T$ is its set of designated service time steps, then at every deterministic T,*

$$\mathbb{E}R_T \leq \min\left\{T, K_T + 4\sqrt{3K_T T\log(2T)} + 1 + \mathbb{E}\sum_{t\in\mathcal{I}_T} \Delta_{I_t}(t)\right\}. \tag{19}$$

Pathwise, $|\mathcal{I}_T| \leq \sum_{i\leq K_T} q_i$.

Hoeffding's inequality bounds the failure at checkpoint i by $\delta/\{2i(i+1)\}$. Summing over potential arm indices gives one coverage event of probability at least $1-\delta/2$, on which every report is correct. For (19), a separate, summable concentration argument at physical time t controls ordinary UCB pulls. Each arm and pre-pull count is charged at most once; inserted service observations only skip counts. This proof holds on all reporting paths and therefore incurs no omitted δT term. Complete proofs appear in the supplementary material.

This separate construction addresses the reporting objective and is not a reimplementation of the two-stage screening algorithm in Section 3. It has no prescribed terminal round, no delayed admission, and no restart that is completed artificially for analysis. The guarantees above concern CP-UCB-AA. The regret guarantee for UCB-AA is stated separately in Section 3.

6 The burden of continual certification

6.1 One uncertain arrival can obstruct many reports

The first lower bound retains one reward environment throughout the time axis. It therefore measures a cumulative burden without adding the worst cases of incompatible experiments. Fix $0 < \varepsilon \leq 1/16$, $0 < \delta \leq 1/16$, and $0 < \eta < \varepsilon/2$. An initial reference has Bernoulli mean $1/4 + \varepsilon/2$. Candidates arrive on any prescribed schedule. In environment zero each has mean $p = 1/4$; environment j changes only candidate j to mean $p + 2\varepsilon$. All candidates are good in environment zero, with boundary distance $\varepsilon/2$. After candidate j arrives in environment j, every other candidate is bad.

Let M_t count arrived candidates and $a_t = M_t - M_{t-1}$. Define

$$\begin{aligned} q_- &= \left\lfloor \frac{\mathrm{kl}(1/4, \delta)}{\mathrm{kl}(1/4, 1/4 + 2\varepsilon)} \right\rfloor, \qquad\qquad Q_t = (Q_{t-1} + q_- a_t - 1)_+, \quad Q_0 = 0, \\ \mathcal{B}_T &= \sum_{t \leq T} (M_t - 1)_+ \mathbf{1}\{Q_t > 0\}. \end{aligned} \tag{20}$$

The queue is an information-capacity certificate: each arriving candidate requires q_- units, while only one observation is possible per round. It is not the queue of any particular policy. In the stated range $q_- \geq 2$.

Theorem 6.1 (Broadcast uncertainty)**.** *For every policy reliable on environment zero and all its one-candidate alternatives,*

$$\mathbb{E}_0 \mathcal{U}_T(\eta) \geq (3/4 - \delta) \mathcal{B}_T.$$

If $\mathcal{B}_T > 0$*, then*

$$\mathbb{P}_0\{\mathcal{U}_T(\eta) \geq \mathcal{B}_T/4\} \geq 1/4.$$

These conclusions still hold when the policy knows the full arrival schedule, the horizon, and the reference distribution.

At a busy time, some deterministic candidate label has expected sample count below q_-. Changing that candidate alone has too little history KL to permit positive certification of another candidate with probability at least 1/4. Reliability also bounds incorrect negative reports. The remaining probability must be unresolved. The same reference environment supports this argument at every time; the difficult label is selected from expected counts rather than from a favorable realized history. The supplementary proof includes the bounded-variable step needed for the probability statement.

If m candidates have arrived by time s, the queue obeys the deterministic inequality

$$\sum_{t \leq s + qm - 1} (M_t - 1)_+ \mathbf{1}\{Q_t(q) > 0\} \geq \frac{q-1}{2} m(m-1), \qquad q \geq 2. \tag{21}$$

The factor m^2 reflects the number of affected reports as well as the work needed to exclude high-quality alternatives. The subtraction of one unit in $q - 1$ accounts for the last observation of each busy period: reports are issued after service.

Corollary 6.2 (Arrival-growth boundary)**.** *Fix the preceding task parameters and one infinite arrival schedule. If a policy is reliable for every reward environment and has a deterministic envelope* $a_T = o(T)$ *satisfying* (14) *in every environment, then* $K_t = o(\sqrt{t})$*. Conversely,* $K_T^2 \log(eK_T) = o(T)$ *suffices for one horizon-free policy to have both* $\mathbb{E}R_T = o(T)$ *and such a sublinear area envelope. In particular, for* $K_t \asymp t^\alpha$, $0 < \alpha < 1$, *simultaneous attainment is possible if and only if* $\alpha < 1/2$.

The sufficiency follows from the next theorem. The exponent statement leaves a logarithmic gap for general critical schedules. It does not identify $K_t = o(\sqrt{t})$ as a sufficient condition on every schedule.

6.2 Worst-schedule maintenance area

Let $\mathfrak{A}_{T,K}$ be the smallest a attainable by a full-model reliable policy with

$$\sup_{\boldsymbol{\tau}: K_T \le K} \sup_{\boldsymbol{r}} \mathbb{P}_{\boldsymbol{r}}\{E_{\text{lab}}^c \text{ or } \mathcal{U}_T(\eta) > a\} \le \delta.$$

Policies may know T, K and the schedule for the lower bound. The upper bound uses neither T nor K as input.

Theorem 6.3 (Worst-schedule area and joint attainability)**.** *For* $K \ge 2$, $T \ge 1$, $0 < \varepsilon \le 1/8$, $0 < \eta \le \varepsilon/32$, *and* $0 < \delta \le 1/16$,

$$2^{-24} \min\left\{KT, \frac{K^2}{\eta^2} \log \frac{K}{\delta}\right\} \le \mathfrak{A}_{T,K} \le 256 \min\left\{KT, \frac{K^2}{\eta^2} \log \frac{K}{\delta}\right\}. \tag{22}$$

The CP policy with $\rho = \eta/4$ *attains the upper order and also satisfies*

$$\mathbb{E}R_T \le \min\left\{T, K + 4\sqrt{3KT\log(2T)} + 1 + 99K\eta^{-2}\log(K/\delta)\right\}. \tag{23}$$

For the upper bound, completed checkpoints imply endpoint errors at most $\eta/4$, sufficient to classify all arms outside the boundary band. The unfinished-work queue is dominated by adding q_i at each arrival and serving at even rounds. On the simultaneous coverage event,

$$\mathcal{U}_T(\eta) \le \min\left\{K_T T, 2K_T \sum_{i \le K_T} q_i\right\}. \tag{24}$$

Additional UCB observations reduce the work; they are not charged again as new checkpoint samples. The lower bound uses $K-1$ candidates with independent means $1/2 \pm 2\eta$ and a reference of mean $1/2+\varepsilon$, all available at the first pull. Simultaneous partial recovery and a horizon truncation give the probability statement required by the definition of $\mathfrak{A}_{T,K}$. The statistical $\log K$ factor has fixed-arm multiple-testing precedents; the lower construction itself requires no repeated arrivals.

For fixed task parameters, (22) implies a precise uniform terminal-count statement: over all schedules with $K_T \le k(T)$, sublinear expected regret and sublinear service envelopes are jointly attainable if and only if $k(T)^2 \log k(T) = o(T)$. Here the worst schedule can change with T. On a single fixed schedule that conclusion cannot be inferred from total growth alone. For example, batches of size

$$m_n = \left\lfloor \frac{\sqrt{s_n}}{(\log s_n)^{1/4}} \right\rfloor, \qquad s_n = 2^{2^n},$$

give $K_t = o(\sqrt{t})$ but preclude any $o(T)$ area envelope uniform over reward environments. Near s_n, simultaneous classification of that burst requires accumulated unresolved area of order $m_n^2 \log m_n$, which exceeds s_n by a growing factor. The supplementary argument selects a single infinite schedule; the difficult reward environment is allowed to vary with the endpoint, as required by the stated uniform guarantee.

7 Temporary membership questions and their regret cost

7.1 A repeated-update family

We now keep the reward environment, arrival schedule, endpoint, and reporting parameters fixed when comparing two area budgets. Choose integers $J \geq 1$, $M \geq \max\{2, J\}$ and

$$0 < \varepsilon \leq 1/8, \quad 0 < \delta \leq 1/16, \quad Jv \leq \varepsilon/4, \quad 0 < d \leq v/32, \quad \eta = d/2. \tag{25}$$

At $a_e = 1 + eH$, $e = 0, \ldots, J$, a reference with mean $b_e = 1/2 + ev$ arrives. For each $e < J$, the same batch contains M candidates with Bernoulli means

$$r_{e,k} = b_e - \varepsilon + \sigma_{e,k} d, \qquad \sigma_{e,k} \in \{-1, +1\}. \tag{26}$$

The last batch contains only its reference, and $T = (J+1)H$. The family Θ includes every sign configuration and every within-batch permutation. The policy is not told which arm is the reference. Lower bounds may disclose those identities and the schedule. All arms remain available.

In its birth window a candidate is good exactly when its sign is positive. Its regret gap is approximately ε, but its distance to the membership boundary is d. At the next update both signs become bad, with boundary distance at least $v - d$. Every reference remains good. Thus the final good set consists only of references, regardless of all the candidate signs. The statistical tasks that distinguish the environments are temporarily relevant and can expire while their arms remain available.

7.2 Area forces completion of enough temporary tasks

The lower-bound ingredient permits arbitrary interleaving and stopping. For m independent uniform signs with observation laws P_i^+, P_i^-, put

$$\kappa = \max_i \max\{D(P_i^+ \| P_i^-), D(P_i^- \| P_i^+)\} \in (0, \infty).$$

If a bounded stopping time has at least half the signs determined with prior probability at least 1/2, and all configurations have familywise report error at most $\delta \le 1/16$, then

$$\overline{\mathbb{E}} \sum_{i=1}^{m} N_i \ge \frac{m}{1024\kappa} \log \frac{m}{\delta}. \tag{27}$$

Other observations must not encode these signs. The proof factors the posterior after the sampling and stopping kernels cancel, and combines a simultaneous recovery bound with coordinatewise change of measure (Kaufmann et al., 2016). It does not assert independence of the random sample counts. The logarithmic simultaneous-recovery scale is established statistical structure; its role here is to price partially completed, expiring tasks.

For clarity, a heterogeneous version gives an explicit lower allocation constraint. Let a compatible batch e have M_e candidates, window length H_e, boundary distance $d_e > \eta$, and two-way KL bounded by κ_e. Put $w_e = M_e H_e$, $V = \sum_e w_e$. Define

$$\Phi_\delta(u) = \begin{cases} \mathrm{kl}(1-\delta-u, \delta), & u < 1-2\delta, \\ 0, & u \ge 1-2\delta, \end{cases}$$

$$g_{m,\delta}(p) = \max\left\{0, \frac{(p-2\delta)_+}{2}\left[\log \frac{m}{4\log 2}\right]_+ - 1, \Phi_\delta(1-p/2)\right\}.$$

If a policy satisfies the joint area requirement with budget a, then

$$\sup_\sigma \mathbb{E}_\sigma R_T \ge \inf_{\substack{0 \le p_e \le 1 \\ \sum_e w_e p_e \ge (1-\delta)V - 2a}} \sum_e \frac{(\varepsilon - d_e) M_e}{\kappa_e} g_{M_e,\delta}(p_e). \tag{28}$$

Here p_e is the probability of determining at least half of batch e before its window ends. A window that never does so contributes at least $w_e/2$ area. This gives the constraint directly from the area success event, with its failure probability retained. The objective then uses (27) under one compatible product prior. The policy can choose where to defer; no separate requirement forces every window to finish. The relaxation need not be attainable in general.

7.3 Reusable coarse evidence and a matched rapid endpoint

CP-UCB-AA can use a predetermined precision ρ_e for each arriving batch. Let $S_e = \sum_{i:\tau_i = a_e} q_i(\rho_e)$ be the total checkpoint work of batch e, and set $M_{-1} = M_J = 0$. In a heterogeneous repeated-update family, suppose

$$H_e \ge 2S_e, \qquad b_J - b_0 \le \varepsilon/4, \qquad d_e + \rho_e + \rho_{e+1} < b_{e+1} - b_e, \qquad 2\max_e \rho_e \le \varepsilon/2. \tag{29}$$

These are conditions on the performance family, not future inputs supplied to the policy. Write $\mathcal{F} = \{e < J : 2\rho_e < d_e\}$.

Proposition 7.1 (Shared update from lifetime evidence)**.** *On the checkpoint coverage event, under*

(29),

$$\mathcal{U}_T(\eta) \le \sum_{e \notin \mathcal{F}} M_e H_e + 2 \sum_{e \in \mathcal{F}} M_e S_e + 2 \sum_{e=0}^{J} (e + 1 + M_{e-1}) S_e. \tag{30}$$

When the next batch completes its checkpoints, every preceding-batch candidate can be excluded using its saved interval and the new reference lower bound, without another observation of that old candidate for certification.

The last term charges only temporarily affected references and the preceding candidate cohort, rather than all historical candidates at each update. The conclusion follows because the new lower bound exceeds every preceding candidate upper bound by more than ε. The condition $H_e \ge 2S_e$ ensures that all its checkpoint observations occur in its birth window. The corresponding unconditional reward bound, including excess UCB pulls, is given in the supplement.

For the homogeneous family (25)–(26), set

$$K = J(M+1)+1, \quad L_K = \log \frac{4K(K+1)}{\delta}, \quad \rho_f = d/4, \quad \rho_c = v/8,$$
$$q_f = \lceil 32 L_K / d^2 \rceil, \quad q_c = \lceil 128 L_K / v^2 \rceil, \quad H = 128(M+J+1) q_f, \quad V = JMH. \tag{31}$$

The quantities q_f, q_c are uniform analysis bounds on the actual armwise quotas. Neither final arm count nor future times are policy inputs. Fine precision can be set from the task input as $\rho_f = \eta/2$; the coarse precision is a separate preselected configuration. Automatic adaptation to an unknown improvement scale is not asserted.

Theorem 7.2 (Rapid and deferred maintenance at the same endpoint)**.** *On the preceding family and horizon,*

$$2^{-20} \frac{\varepsilon J M}{d^2} \log \frac{M}{\delta} \le \mathfrak{R}_\Theta(V/8) \le 2^{18} \frac{\varepsilon J M}{d^2} \log \frac{M}{\delta}. \tag{32}$$

The coarse-precision CP policy has joint label-and-area success probability at least $1-\delta$ with area budget $9V/8$, and

$$\sup_{\theta \in \Theta} \mathbb{E}_\theta R_T \le 2\varepsilon J M q_c + 64 \left(\frac{JM}{\varepsilon} + \frac{J^2}{v} \right) \log(2T) + 1. \tag{33}$$

Consequently, at fixed legal $J, M, \varepsilon, v, \delta$ and $d \downarrow 0$, with $\eta = d/2$ and the horizon in (31), $\mathfrak{R}_\Theta(V/8) = \Theta(d^{-2})$, whereas $\mathfrak{R}_\Theta(9V/8) = O(\log(1/d))$.

For the lower bound, area at most $V/8$ forces half-completion in at least $3J/4$ windows on success. Enough windows then require (27), and each candidate observation costs at least $\varepsilon - d$. Distinct batches' samples are charged once. For the upper bound, (30) gives the appropriate area budgets. At the specified horizon the deterministic bound $R_T \le 2\varepsilon T$ suffices for the rapid order; it would not justify the same conclusion at arbitrarily longer windows. The coarse bound instead counts checkpoint observations and excess ordinary UCB pulls separately.

The two budgets differ by a constant factor, but they allow different service levels. The theorem does not claim that the coarse policy meets the rapid budget, that its regret upper bound is matched, or that all intermediate budgets are characterized. A preselected mixture of fine and coarse batches gives additional attainable points, as recorded in the supplement.

7.4 A probability lower bound under the same area requirement

An expectation lower bound alone could, in principle, reflect unusual sample paths. The following result gives a stronger conclusion without replacing the cumulative area requirement by deadlines.

Theorem 7.3 (A nonvanishing probability of the rapid-maintenance cost)**.** *In the repeated-update family, suppose a reliable policy satisfies*

$$\mathbb{P}_\sigma\{E_{\mathrm{lab}}, \mathcal{U}_T(\eta) \le V/8\} \ge 1-\delta$$

for every sign configuration. Then one fixed configuration satisfies

$$\mathbb{P}_\sigma\left\{R_T \ge 2^{-20}\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta}\right\} \ge \frac{11}{80}. \tag{34}$$

The proof stops each window at half-completion, its endpoint, or the first candidate sample beyond a fixed budget. Applying the partial-recovery lemma to this stopped experiment bounds the probability of a cheap completion by $1/2$. Low total area requires many completions; a count of cheap completions then leaves enough expensive windows with probability at least $1/5-\delta \ge 11/80$. Their actual observations give (34). No independence between window-completion events is needed. Since $11/80 > \delta$, the rapid endpoint in (32) also has matching order when regret and area are bounded on one event of probability at least $1-\delta$. The full stopping argument and a compatible multiple-window area truncation are proved in the supplement.

8 Numerical studies

8.1 The original two arrival scenarios

We retain the original simulation designs and their three evaluation criteria: cumulative dynamic regret R_T, wasted pulls

$$W_T = \sum_{t=1}^{T} \mathbf{1}\{\Delta_{I_t}(t) \ge 0.1\},$$

and active-set size. The latter counts arms retained by an implementation; it does not by itself establish that every available arm has been evaluated. Both archived scenarios have $T = 60{,}000$ and 100 independent replications.

The progressive step-up scenario starts with ten arms. Background arrivals occur with Bernoulli probability 0.005 per time step. Step arms arrive at $t/T \in \{0.3, 0.6, 0.9\}$ with means $\min(1, 0.7+0.05k)$, $k = 1, 2, 3$. The near-tied top-set scenario starts with twenty arms, including five arms drawn from $\mathrm{Uniform}(0.75, 0.90)$. Background arrivals again have probability 0.005, with probability 0.30 of being drawn from the near-optimal component. These are the archived configurations that generated the original figures; the supplied configuration files specify the remaining mixture components.

Table 1: Original archived experiments at $T = 60{,}000$ (100 replications per scenario and method). Entries are means with Monte Carlo standard errors in parentheses. Archived UCB-AA is the explore-then-commit implementation with $M = 5$ in C and $M = 4$ in E; these are not validation results for a repaired or continuously admitting implementation. Buffered arrivals excluded from the final active count were not necessarily screened or certified inferior.

Scenario	Method	R_T	W_T (0.1)	Final active
C	Archived UCB-AA	5,139.9 (337.1)	8,225.2 (765.3)	3.2 (0.3)
C	Growing UCB1	9,347.2 (60.1)	16,378.0 (121.2)	318.4 (1.8)
C	Growing MOSS-t	2,036.7 (24.1)	3,331.0 (30.5)	318.4 (1.8)
C	Naive elimination	27,507.0 (198.2)	50,276.7 (395.6)	2.6 (0.1)
C	Dynamic SE	28,872.7 (107.6)	52,995.8 (141.9)	87.0 (2.1)
E	Archived UCB-AA	1,880.9 (78.9)	3,264.1 (127.5)	5.7 (0.2)
E	Growing UCB1	7,964.3 (37.9)	21,281.4 (225.8)	325.4 (1.8)
E	Growing MOSS-t	2,043.8 (20.3)	5,010.2 (55.6)	325.4 (1.8)
E	Naive elimination	5,433.4 (182.4)	9,436.1 (350.5)	5.9 (0.2)
E	Dynamic SE	25,152.2 (98.4)	46,331.8 (152.8)	133.2 (1.5)

The archived UCB-AA implementation uses $M = 5$ and $M = 4$ in the two scenarios, respectively, and $N = e/\widetilde{\Delta}_M^2$. It stops admission after the prescribed rounds and then exploits a fixed survivor pool. Its formal comparisons use incumbent references when present and empirical means from all observations. These details differ from the prefix-based, merged-reference procedure stated in Section 3. We therefore retain these results as observations about the archived implementation, rather than claiming that the repaired finite-budget procedure has been rerun.

The archived UCB-AA has mean regret 5139.94 in the step-up scenario, compared with 2036.72 for Growing MOSS. In the near-tied scenario the corresponding means are 1880.88 and 2043.75. In both scenarios it improves mean regret and wasted pulls relative to the archived Growing UCB1, dynamic successive elimination, and no-pre-elimination implementations. Thus the favorable near-tied result is retained, as is the substantially worse step-up result against MOSS. A small survivor pool cannot be attributed entirely to statistical screening: some later arms were never admitted. The original six figures are retained in the supplement with this implementation qualification.

To evaluate the repaired UCB-AA procedure itself, the two original environments need a common implementation of the prefix means, merged references, budget truncation, and continued admission. Short-interval retained-history comparisons and best-arm-retention diagnostics are also needed. The reporting experiments below answer a different, additional question and do not replace that algorithm comparison.

8.2 Finite-budget comparisons for reliable set maintenance

We examine the cost of maintaining reliable reports at an actual sampling budget, including settings in which the conservative checkpoint construction is impractical. These comparisons assess finite-budget behavior of the reporting construction, not the two-stage UCB-AA procedure. The four environments are not the separated family used for the two-endpoint theorem, and the simulations do not test its asymptotic rates.

8.2.1 Protocol and competing rules

All rewards are Bernoulli. We fix $T = 20{,}000$, $\varepsilon = .30$, $\eta = .10$, and $\delta = .05$, with eight arms by the end of each run. There are 160 independent replications per environment. Within a replication, every method receives the same potential reward stream for each arm; its nth pull of arm i reveals the same $Y_{i,n}$. Thus methods share the underlying observations without being forced to take the same actions. Arrivals occur before the action, reports follow the observation, and all methods retain previous observations. The full arrival times, means, seeds, and policy definitions are saved in a single configuration file.

The first environment has initial means $.65, .50$ and three later batches of inferior arms, so the benchmark remains constant. The second has four successive benchmark levels $.45, .70, .90, .98$; some earlier unresolved membership questions become obsolete when a better arm arrives. The third introduces six arms at one time, including a new best arm. The fourth ends arrivals at pull 4037 and leaves nearly 16,000 further pulls for recovery. Noninitial batches occur 36 pulls after a 100-pull admission grid, so the short-period baseline is not evaluated only at its most favorable alignment. No policy is given future arrivals, the means, or the evaluation horizon.

We implement CP-UCB-AA with FIFO service every second pull at two fixed precisions. The choice $\rho = .04$ satisfies $2\rho < \eta$: on the coverage event, completing all checkpoints resolves every η-separated arm. The choice $\rho = .12$, inherited from the implementation supplied with the theoretical development, is a coarser engineering setting. Its labels remain reliable, but it need not resolve every arm counted by the $\eta = .10$ metric. We report these choices separately. With eight arms their total checkpoint quotas are 73,083 and 8,123 pulls, respectively. The finer setting therefore cannot complete all quotas within the chosen budget.

Three comparisons isolate the allocation rule while retaining the coarser checkpoint report: FIFO service every eighth pull, confidence-driven service every second pull, and full-history UCB without designated service pulls. Confidence-driven service samples the widest interval among unresolved arms and the arm with largest upper endpoint, breaking ties by sample count and then arrival order. It stops designated service when no report remains unresolved. A completed checkpoint retains its fixed interval.

We also use the full-prefix confidence report from the theory with three reward or sampling rules. UCB-prefix admits every arm immediately and uses the same full-history UCB index as CP-UCB-AA. Growing-MOSS uses the anytime index

$$\hat{r}_i(n) + \sqrt{\frac{\max\{\log(t/(K_t n)), 0\}}{n}},$$

with infinite index at $n = 0$; no minimax property of the fixed-horizon MOSS algorithm is assumed for this extension. ALL-ε adaptation uses the confidence-driven service rule every second pull and UCB otherwise. It retains all history and can revise positive reports after arrivals. This is a natural adaptation of confidence-based all-good-arm sampling, not an implementation of FAREAST or $(\mathrm{ST})^2$. Finally, Batch100-UCB follows the reviewer's proposed short-admission comparison: it admits waiting arms at pulls $1, 101, 201, \ldots$, uses UCB on the admitted reward-active arms, and

Table 2: Final cumulative dynamic regret and cumulative η-unresolved arm-time at $T = 20{,}000$. Entries are means (Monte Carlo standard errors), from 160 paired replications. Area is in units of 10^3. The two checkpoint precisions have different completeness guarantees.

Scenario	Method	Regret	Area / 10^3
Stable benchmark	CP-FIFO ($\rho = .04$)	1360.5 (1.5)	111.78 (0.00)
	CP-FIFO ($\rho = .12$)	3629.9 (0.7)	49.72 (0.23)
	UCB-prefix	441.7 (2.3)	83.97 (0.46)
	Growing-MOSS	77.3 (0.8)	92.43 (0.00)
	ALL-ε adapt.	1367.8 (11.6)	11.31 (0.09)
	Batch100-UCB	209.3 (2.6)	85.27 (0.52)
Successive improvements	CP-FIFO ($\rho = .04$)	5297.4 (1.2)	74.82 (0.00)
	CP-FIFO ($\rho = .12$)	1194.7 (0.0)	14.74 (0.10)
	UCB-prefix	246.2 (1.7)	18.45 (0.14)
	Growing-MOSS	53.1 (0.8)	33.65 (0.11)
	ALL-ε adapt.	2905.8 (2.4)	9.53 (0.13)
	Batch100-UCB	238.8 (2.1)	22.85 (0.09)
Burst	CP-FIFO ($\rho = .04$)	2798.0 (1.6)	75.89 (0.00)
	CP-FIFO ($\rho = .12$)	3392.2 (0.2)	28.21 (0.17)
	UCB-prefix	354.1 (2.2)	41.75 (0.39)
	Growing-MOSS	69.3 (0.9)	57.03 (0.01)
	ALL-ε adapt.	2017.4 (9.8)	10.31 (0.09)
	Batch100-UCB	278.9 (2.7)	55.87 (0.15)
Arrivals stop	CP-FIFO ($\rho = .04$)	4759.0 (1.3)	89.86 (0.00)
	CP-FIFO ($\rho = .12$)	2645.9 (0.3)	25.14 (0.01)
	UCB-prefix	248.8 (1.5)	45.96 (0.56)
	Growing-MOSS	52.0 (0.6)	71.59 (0.01)
	ALL-ε adapt.	3027.5 (4.3)	12.07 (0.07)
	Batch100-UCB	214.2 (1.8)	55.28 (0.21)

retires an arm from reward sampling when its upper bound falls below the largest active lower bound. It retains all observations. Its report covers all physically arrived arms, including those still waiting for admission, whose intervals are $[0, 1]$.

Table 2 gives means and Monte Carlo standard errors. Figure 2 includes every method; error bars are 1.96 Monte Carlo standard errors. The released data also contain paired differences, all unresolved paths, service pulls, and the regret charged to service. The compiled implementation was checked against the original CP-UCB-AA `choose/observe/report` class for 20,000 successive rounds: actions, service indicators, every available-arm count, and every reported label agreed at each round.

8.2.2 What the comparisons show

In all 640 finer-checkpoint runs, every available arm remains unresolved at every time step. Five to eight arms are counted by the final η-unresolved metric because that metric exempts the boundary band; the remaining arms have not been classified. The resulting zero-error record therefore does not establish useful classification at this budget. Its charged unresolved area ranges from 74,820 to

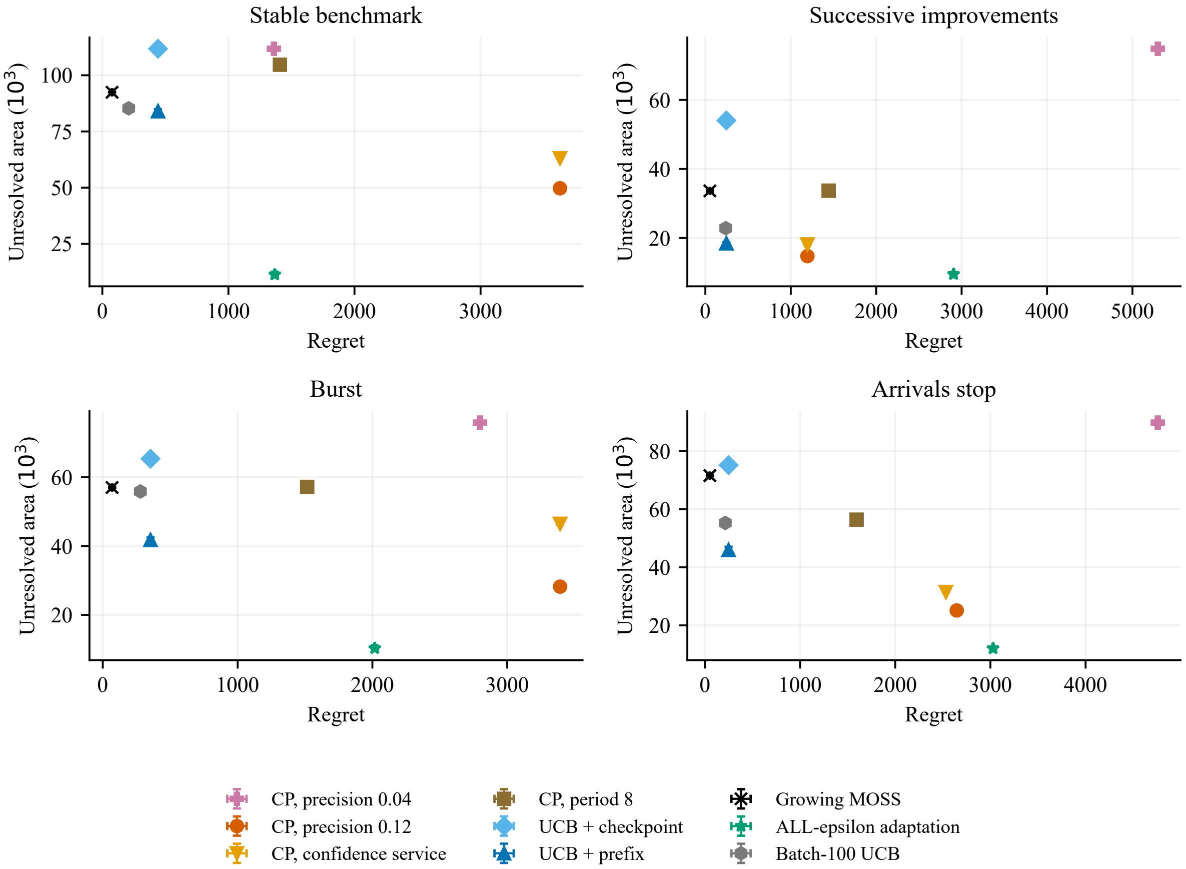

Figure 2: Regret and cumulative η-unresolved area for all nine methods. Lower values on both axes are preferable. Error bars quantify Monte Carlo uncertainty, not uncertainty in the theoretical reliability statement. Neither checkpoint precision nor its allocation variants uniformly dominate the confidence-based adaptation.

111,784. This is an observable limitation of the conservative construction, rather than evidence that arrivals are intrinsically unmanageable at this budget. The confidence-based adaptation resolves all η-separated arms by the end in every replication of these four environments.

The coarser checkpoints reduce area substantially relative to UCB alone, but can incur much larger regret. Under a stable benchmark, CP-FIFO has mean regret 3629.9 and area 49,718, compared with 1367.8 and 11,306 for ALL-ε adaptation. The paired differences, CP-FIFO minus the adaptation, are 2262.1 (standard error 11.7) for regret and 38,411.7 (standard error 241.9) for area. The adaptation also improves both quantities in the burst environment. These comparisons do not support a general numerical advantage of checkpoint FIFO over a strong history-retaining identification rule.

In the successive-improvement environment, CP-FIFO instead occupies a different cost point: regret 1194.7 and area 14,743, compared with 2905.8 and 9525 for the confidence-based adaptation. Here the paired regret difference is -1711.0 (standard error 2.4), while the area difference is 5217.7 (standard error 141.1). These points do not isolate the benefit of task expiration. The adaptation

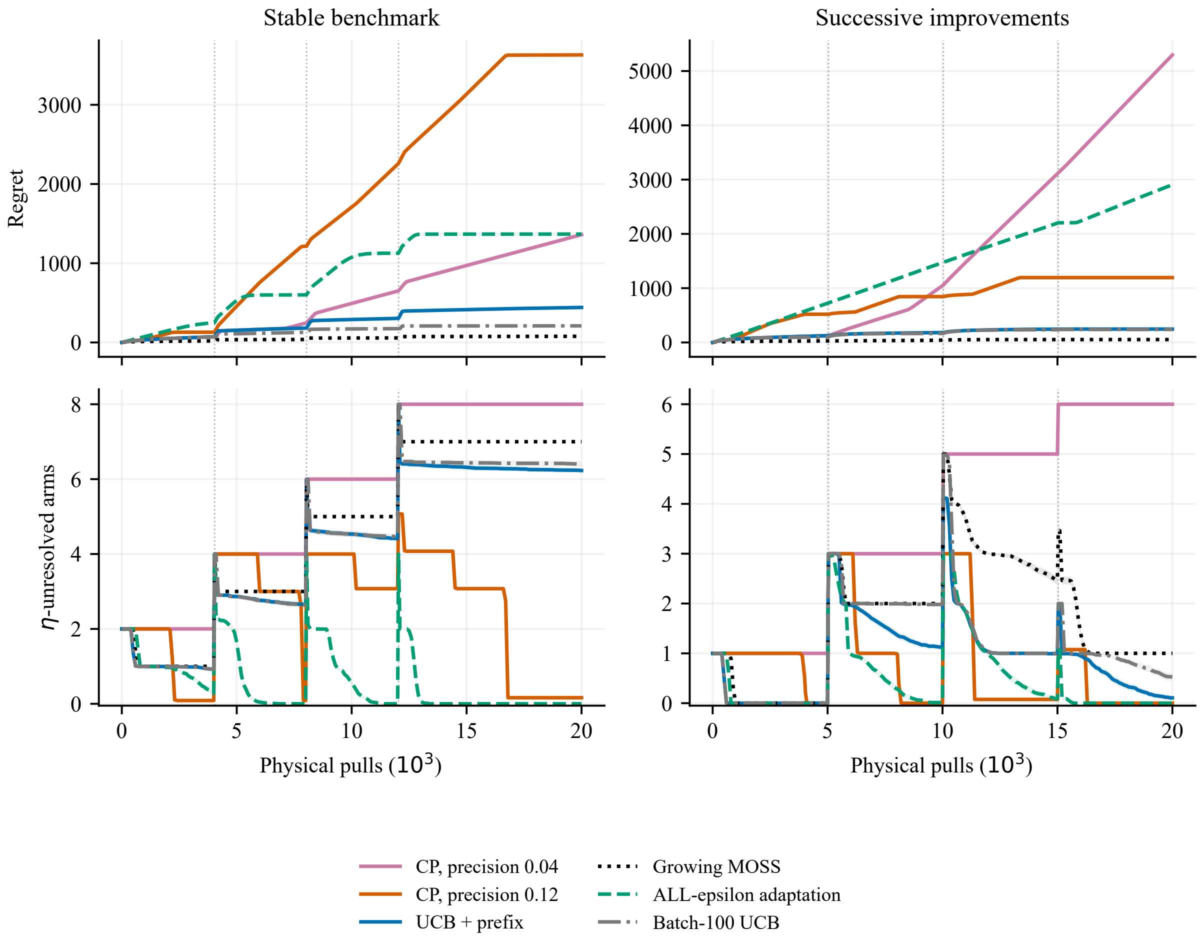

Figure 3: Mean regret and mean η-unresolved arm counts at the physical sampling clock. Bands are 1.96 Monte Carlo standard errors; vertical dotted lines mark arrivals. Stable and improving benchmarks are shown here; the other environments appear in the supplement. The main six methods are displayed; all nine methods and all sampled paths are retained in the released results.

uses all 10,000 designated service pulls in this environment, while coarse CP uses about 5,786; some remaining questions lie inside the unpenalized boundary band. The lower regret may therefore partly reflect a different stopping rule for reporting work. A comparison with the same boundary tolerance and service requirement is needed to distinguish the mechanisms. These observations do not estimate an optimal frontier.

Growing-MOSS has the smallest mean regret in all four environments, between 52.0 and 77.3, while leaving substantially larger unresolved areas. Low regret therefore provides little evidence of timely set maintenance. Conversely, the explicit service rules pay for observations of inferior arms. Reducing service from every second to every eighth pull generally reduces regret and enlarges unresolved area. Replacing FIFO by the confidence-driven checkpoint allocation does not uniformly improve either criterion; allocation and the report's precision both matter.

For Batch100-UCB we additionally record the exact decomposition into within-active regret, admission delay, and loss from previous reward retirement. Its admission-delay contributions are 0, 33.92, 16.00, and 22.40 in the four environments. No historical-retirement deficit occurred in these runs. The largest discrepancy between total regret and the sum of the three recorded components was below 3.1×10^{-12}. The corresponding admission and retirement terms are identically zero for the methods that admit every arm immediately and retain every arm for reward selection. This accounting avoids attributing every loss outside within-active learning to arrival delay.

There were no incorrect positive or negative reports in any of the 5760 runs. For one fixed environment and method, zero failures in 160 replications gives a one-sided 95% upper confidence bound of 0.0186 on its failure probability. This finite set of diagnostics does not establish the uniform, all-environment reliability guarantee. Nor do these small eight-arm experiments validate the asymptotic lower bounds, resolve the critical logarithmic gap, or establish checkpoint optimality. They show the practical cost of the conservative quotas and the importance of comparing retained-history methods under both regret and maintenance criteria.

9 Discussion

We studied stochastic multi-armed bandits with arriving arms, where the available action set expands and performance is measured against the best currently available arm. Arrival information discrepancy and a drifting benchmark create repeated learning costs even though every arm has a stationary reward law. The two-stage UCB-AA design addresses the information imbalance by screening arrivals before full competition with incumbents. The revised regret account separates learning, admission delay, and loss of an earlier admitted benchmark. Its finite-budget guarantee is conditional on the stated calibration and retains the admission-delay term explicitly.

The additional maintenance analysis makes precise a statistical question that the original active-set comparison left open. A smaller internal set does not establish reliable knowledge of the complete near-optimal set. One uncertain arrival can obstruct many earlier certifications, and a drifting benchmark can make difficult unresolved questions expire. The broadcast and repeated-update results characterize costs arising from these two effects. Their assumptions, including the restricted update family and the different allowed area budgets, are part of the conclusions. The worst area lower bound already occurs in a static initial batch; the temporary-task results supply a separate dynamic mechanism.

UCB-AA and CP-UCB-AA serve different purposes in this analysis. The former retains the original screening architecture. The latter is a constructive reporting rule whose service precision is fixed in advance and whose unconditional reward analysis does not require correct reports. Its conservative checkpoints may be ineffective at moderate budgets, and its numerical cost is not uniformly competitive. Neither a negative report nor a smaller reported set is treated as a reduction in total storage.

The supplement supplies full proofs of the quantile lower bound in Section 7 and the compatible-window truncation result. It also gives eventual exact classification by a separate revisable multiresolution construction when arrivals stop, the final arm count is finite, and no arm lies on the membership boundary. That conclusion does not apply to fixed-precision checkpoints.

The general frontier under overlapping arrivals, automatic precision selection, and a matching converse for deferred maintenance remain open. The archived UCB-AA study also leaves a concrete empirical obligation: the repaired procedure must be compared under the same admission and retained-history rules before attributing a numerical gain to pre-elimination. Contextual information, delayed feedback, and post-arrival reward drift remain natural extensions of the original MAB-AA formulation.

Data and code availability

No original empirical or human-subject outcome data were collected. The original simulation configurations, endpoint records, source snapshots, and figures are preserved separately from the new reporting experiments. The release contains the 5,760 reporting runs, their fixed configuration, code, and checks. The oncology illustration is the historical motivation from the original submission, based on public approval records.

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